

Electrophoresis is a laboratory method for separating molecules based on size, electric charge, binding affinity, or other properties. Many important biological molecules—including proteins, DNA, and RNA—have different sizes or charges based on their biological purpose. For electrophoresis, samples are loaded into sample wells and the voltage source generates a uniform electric field across a gel between the anode and cathode (Figure 1). The molecules in the samples experience an electrostatic force, causing them to migrate along the lanes in the gel. Molecules like DNA and RNA are typically negatively charged, and thus migrate toward the anode.

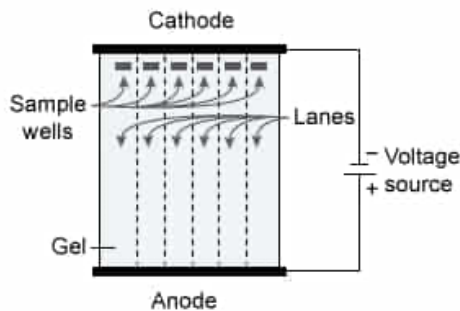


Figure 1 Schematic of an electrophoresis device

Electrophoresis sorts samples based on charge when the gel does not restrict migration of the molecules. Alternatively, samples are sorted by size when the gel impedes the migration of larger molecules, as illustrated in Figure 2.

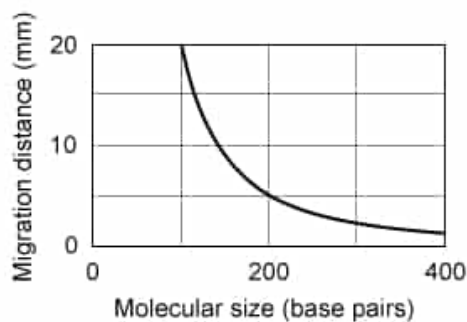


Figure 2 Plot of migration distance for molecules with different sizes

After sufficient time has elapsed for clear separation of the migration distances for each sample, the sample positions in the gel are detected by applying a stain or by tagging the molecules with markers that produce fluorescent or radioactive signals.

A 100 μL sample of an unknown protein has a mass of 130 mg. What is the density of the protein sample? (Note: 1 L = 1,000 cm^3 .)

- ☐ A. 0.77 kg/m^3 [1%]
- ☐ B. 1.3 kg/m^3 [46%]
- ☐ C. 1,000 kg/m^3 [1%]
- ☒ D. 1,300 kg/m^3 [50%]

Correct

50% answered correctly

Explanation:

Density of a protein sample

$$\rho = \frac{m}{V}$$

- ρ Density
- m Mass
- V Volume



$$\rho = \frac{130 \text{ mg}}{100 \text{ }\mu\text{L}} = 1.3 \frac{\text{kg}}{\text{L}} = 1300 \frac{\text{kg}}{\text{m}^3}$$

$$\rho = 1300 \frac{\text{kg}}{\text{m}^3}$$

Sample

$m = 130 \text{ mg}$, $V = 100 \text{ }\mu\text{L}$

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The **density** ρ of a material is the ratio of its **mass** m and its **volume** V :

$$\rho = \frac{m}{V}$$

In this question, a 100 μL sample has a mass of 130 mg. Substituting these values into the equation for ρ yields:

$$\rho = \frac{130 \text{ mg}}{100 \text{ }\mu\text{L}}$$

Converting the units of mg to kg and μL to L gives:

$$\rho = \frac{(130 \text{ mg}) \left(\frac{10^{-3} \text{ g}}{1 \text{ mg}} \right) \left(\frac{10^{-3} \text{ kg}}{1 \text{ g}} \right)}{(100 \text{ }\mu\text{L}) \left(\frac{10^{-6} \text{ L}}{1 \text{ }\mu\text{L}} \right)}$$

$$\rho = 1.3 \frac{\text{kg}}{\text{L}}$$

The units of ρ should be kg/m^3 . Because $1 \text{ L} = 1,000 \text{ cm}^3$, there are 10^{-3} L for each 1 cm^3 . From the relationship $1 \text{ m} = 100 \text{ cm}$, cubing both sides gives $1 \text{ m}^3 = 100^3 \text{ cm}^3 = 10^6 \text{ cm}^3$. Therefore, converting ρ to the proper units yields:

$$\rho = \left(1.3 \frac{\text{kg}}{\text{L}} \right) \left(\frac{10^{-3} \text{ L}}{1 \text{ cm}^3} \right) \left(\frac{10^6 \text{ cm}^3}{1 \text{ m}^3} \right)$$

$$\rho = 1,300 \frac{\text{kg}}{\text{m}^3}$$

(Choice A) $0.77 \text{ kg}/\text{m}^3$ is obtained by incorrectly dividing V by m .

(Choice B) $1.3 \text{ kg}/\text{m}^3$ is obtained by not correctly converting the units from cm^3 to m^3 .

(Choice C) $1,000 \text{ kg}/\text{m}^3$ is obtained by incorrectly using 100 mg for m .

Educational objective:

The density of a material is equal to the ratio of its mass and its volume.

Time spent: 0 sec

Subject:
Physics

QID:403458

System:
Fluids

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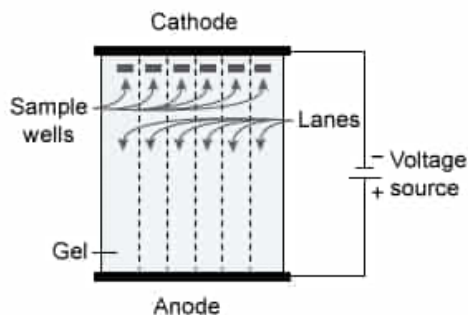


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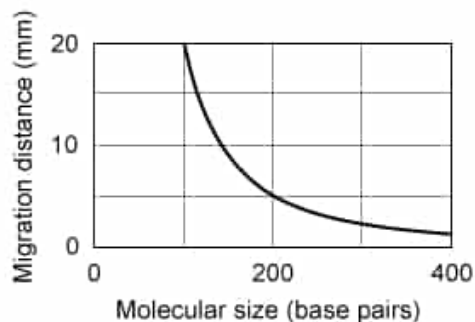


Figure 2 Plot of migration distance for molecules with different sizes

After sufficient time has elapsed for clear separation of the migration distances for each sample, the sample positions in the gel are detected by applying a stain or by tagging the molecules with markers that produce fluorescent or radioactive signals.

During electrophoresis, a DNA molecule is stretched 0.8×10^{-6} m beyond its normal unstretched length, giving it 0.1×10^{-18} J of elastic potential energy. What is the approximate elastic constant for the DNA molecule?

- ☐ A. 3.0×10^{-6} N/m [9%]
- ☒ B. 0.30×10^{-6} N/m [52%]
- ☐ C. 0.15×10^{-6} N/m [17%]
- ☐ D. 0.25×10^{-12} N/m [21%]

Correct

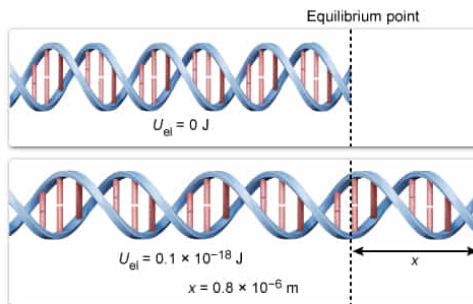
51% answered correctly

Explanation:

Elastic constant of a DNA molecule

$$U_{el} = \frac{1}{2} kx^2$$
$$k = \frac{2U_{el}}{x^2}$$

- U_{el} Elastic potential energy
- k Elastic constant
- x Displacement



$$k = \frac{2(0.1 \times 10^{-18} \text{ J})}{(0.8 \times 10^{-6} \text{ m})^2} \approx 0.3 \times 10^{-6} \frac{\left(\frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}\right)}{\text{m}^2}$$
$$k \approx 0.3 \times 10^{-6} \frac{\text{N}}{\text{m}}$$

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An unstretched spring (ie, at its **equilibrium point**) has zero **elastic potential energy** U_{el} . Stretching or compressing the spring increases its U_{el} because changing its length requires doing **work** against the elastic forces within the spring. Other objects, from rubber bands to individual molecules, can also undergo elongation and compression, increasing U_{el} . The value of U_{el} is equal to half the product of the **elastic constant** k of the material and the square of the displacement x away from the equilibrium point:

$$U_{el} = \frac{1}{2} kx^2$$

This equation can be rearranged to solve for k in terms of U_{el} and x :

$$k = \frac{2U_{el}}{x^2}$$

In this question, U_{el} is given as $0.1 \times 10^{-18} \text{ J}$ and x as $0.8 \times 10^{-6} \text{ m}$. Substituting these values into the equation above yields:

$$k = \frac{2(0.1 \times 10^{-18} \text{ J})}{(0.8 \times 10^{-6} \text{ m})^2}$$
$$k = \frac{0.2 \times 10^{-18} \text{ J}}{0.64 \times 10^{-12} \text{ m}^2}$$
$$k \approx 0.30 \times 10^{-6} \frac{\text{J}}{\text{m}^2}$$

Converting the units for k into standard units (N/m) gives:

$$\frac{\text{J}}{\text{m}^2} = \left(\frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}\right) \left(\frac{1}{\text{m}^2}\right)$$
$$\frac{\text{J}}{\text{m}^2} = \left(\frac{\text{kg} \cdot \text{m}}{\text{s}^2}\right) \left(\frac{1}{\text{m}}\right)$$
$$\frac{\text{J}}{\text{m}^2} = \frac{\text{N}}{\text{m}}$$

Therefore, the approximate value of k is:

$$k \approx 0.30 \times 10^{-6} \frac{\text{N}}{\text{m}}$$

(Choice A) $3.0 \times 10^{-6} \text{ N/m}$ is obtained by incorrectly using $1 \times 10^{-18} \text{ J}$ for U_{el} .

(Choice C) $0.15 \times 10^{-6} \text{ N/m}$ is obtained by not including the $\frac{1}{2}$ term in the equation for U_{el} .

(Choice D) $0.25 \times 10^{-12} \text{ N/m}$ is obtained by not squaring x in the equation for U_{el} .

Educational objective:

Stretching a molecule beyond its equilibrium point imparts elastic potential energy to the molecule. The elastic constant of the molecule is calculated from the elastic potential energy and the displacement from its equilibrium point.

Time spent: 0 sec

Subject:
Physics

QID: 403459

System:
Fluids

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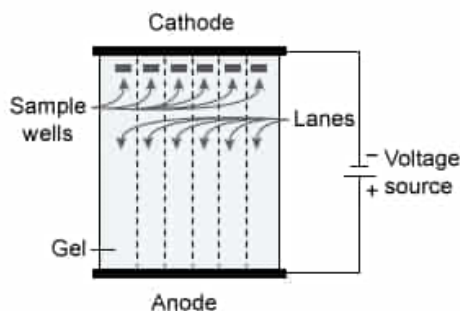


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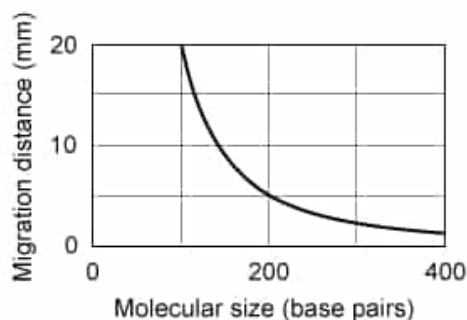


Figure 2 Plot of migration distance for molecules with different sizes

After sufficient time has elapsed for clear separation of the migration distances for each sample, the sample positions in the gel are detected by applying a stain or by tagging the molecules with markers that produce fluorescent or radioactive signals.

Which of the following best describes the mathematical relationship between migration distance d and molecular size s in Figure 2?

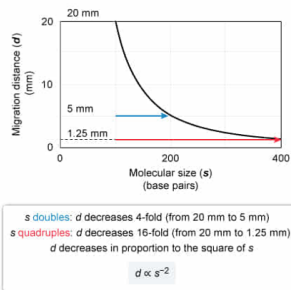
- ☐ A. $d \propto s^2$ [5%]
- ☐ B. $d \propto s$ [1%]
- ☐ C. $d \propto s^{-1}$ [24%]
- ☒ D. $d \propto s^{-2}$ [68%]

Correct

68% answered correctly

Explanation:

Mathematical relationship between migration distance and molecular size



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A graph of two variables provides information about the mathematical relationship between them, including if the variables are **directly proportional**, **inversely proportional** or **linearly related** to each other. Information about the mathematical relationship can also be determined by evaluating how the value of one variable changes when the other variable increases or decreases by specific amounts.

In this question, the graph in Figure 2 shows that d is nonlinear with respect to s (ie, it does not follow a straight line). Furthermore, d is inversely proportional to a power of s because d decreases as s increases. From the graph, doubling the value of s from 100 to 200 results in a four-fold decrease in d , from 20 to 5. In addition, quadrupling the value of s from 100 to 400 results in a sixteen-fold decrease in d , from 20 to 1.25. Based on these changes in d as s increases, d is inversely proportional to the square of s (ie, $d \propto s^{-2}$).

(Choices A and B) These relationships suggest d increases as s increases. However, Figure 2 shows d decreasing at larger values of s .

(Choice C) This relationship suggests that d decreases by a factor of two when s is doubled, and decreases by a factor of four when s is quadrupled. However, Figure 2 shows that d decreases faster than s^{-1} as s increases.

Educational objective:

The mathematical relationship between two variables can be determined from a graph of the two variables.

Time spent: 0 sec

Subject:
Physics

QID: 403460

System:
Fluids

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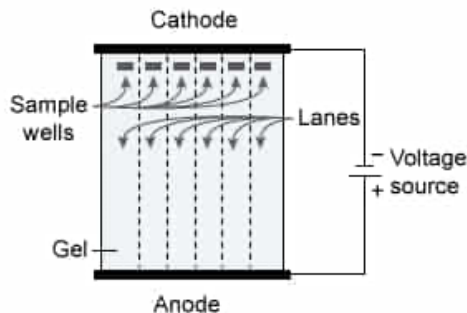


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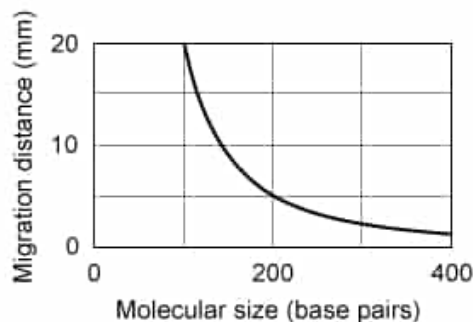


Figure 2 Plot of migration distance for molecules with different sizes

After sufficient time has elapsed for clear separation of the migration distances for each sample, the sample positions in the gel are detected by applying a stain or by tagging the molecules with markers that produce fluorescent or radioactive signals.

Two protein samples are placed in an electrophoresis device. One protein sample contains mostly lysine whereas the other protein contains mostly aspartate. Would both samples accelerate toward the anode when the voltage source of the electrophoresis device is activated?

- ☐ A. No, because the lysine-rich protein is negatively charged and the aspartate-rich protein is positively charged [4%]
- ✓ ☒ B. No, because the lysine-rich protein is positively charged and the aspartate-rich protein is negatively charged [92%]
- ☐ C. Yes, because both proteins are negatively charged [2%]
- ☐ D. Yes, because both proteins are positively charged [1%]

Explanation:

Acceleration of charged proteins during electrophoresis

$\vec{F}_E = q\vec{E}$
 $\vec{a} = \frac{\vec{F}_E}{m}$

- $\vec{F}_E$ Electrostatic force
- q Charge
- $\vec{E}$ Electric field
- $\vec{a}$ Acceleration
- m Mass

Positive Charge

[NH3+]CC(C(=O)[O-])CC

Lysine (K)
Lys

Cathode

Anode

Negative Charge

[O-]C(CC(=O)[O-])CC

Aspartate (D)
Asp

Aspartate-rich protein (negative charge) accelerates towards anode (positive charge)
 Lysine-rich protein (positive charge) accelerates towards cathode (negative charge)

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The **electrostatic force** F_E on an object equals the product of the object's **electric charge** q and the **electric field** E surrounding the object:

$$\vec{F}_E = q \vec{E}$$

E is directed away from the positive electrode and toward the negative electrode. The **sign of q** determines if F_E and E are in the same direction (positive q) or opposite directions (negative q).

Some amino acids are positively charged (eg, arginine, lysine, histidine) whereas others are negatively charged (eg, tyrosine, cysteine, glutamate, aspartate). The overall q of a protein depends on its constituent amino acids. The **acceleration** a of the protein equals the ratio of F_E and the mass m of the protein:

$$\vec{a} = \frac{\vec{F}_E}{m}$$

Substituting the definition of F_E into the equation above yields:

$$\vec{a} = \frac{q \vec{E}}{m}$$

In this question, the electrophoresis anode has a positive q and the cathode has a negative q . The aspartate-rich protein has a negative q and accelerates toward the anode (ie, opposite the direction of E). The lysine-rich protein has a positive q and accelerates toward the cathode. Therefore, both protein

samples would not accelerate toward the anode because only the aspartate-rich protein is negatively charged.

(Choice A) The lysine-rich protein is positively charged and the aspartate-rich protein is negatively charged.

(Choices C and D) Both protein samples would not accelerate toward the anode because they have different values for q .

Educational objective:

The overall charge of a protein depends on the charges of the constituent amino acids. During electrophoresis, negatively charged proteins accelerate toward the positively charged anode.

Time spent: 0 sec
 Subject:
 Physics

QID: 403461
 System:
 Fluids

Electrophoresis is a laboratory method for separating molecules based on size, electric charge, binding affinity, or other properties. Many important biological molecules—including proteins, DNA, and RNA—have different sizes or charges based on their biological purpose. For electrophoresis, samples are loaded into sample wells and the voltage source generates a uniform electric field across a gel between the anode and cathode (Figure 1). The molecules in the samples experience an electrostatic force, causing them to migrate along the lanes in the gel. Molecules like DNA and RNA are typically negatively charged, and thus migrate toward the anode.

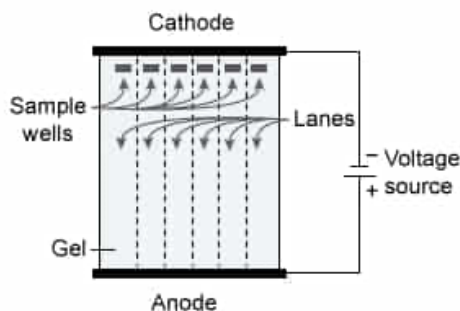


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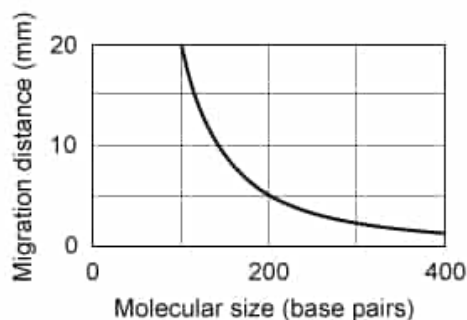


Figure 2 Plot of migration distance for molecules with different sizes

After sufficient time has elapsed for clear separation of the migration distances for each sample, the sample positions in the gel are detected by applying a stain or by tagging the molecules with markers that produce fluorescent or radioactive signals.

A researcher wants to increase the acceleration of protein samples during electrophoresis. Which of the following changes will increase the acceleration of the samples? (Note: Assume that the time the voltage source is on is constant, the samples do not actually reach the anode, and that friction is negligible during protein migration.)

- ☐ A. Increasing the number of protein molecules in the samples [6%]
- ☐ B. Increasing the distance between the anode and the cathode [16%]
- ☒ C. Decreasing the distance between the anode and the cathode [73%]
- ☐ D. Decreasing the voltage [3%]

Correct

72% answered correctly

Explanation:

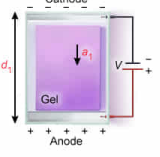
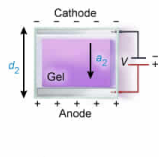
Explanation.

Decreasing anode/cathode distance increases acceleration of samples

$$F_E = ma \quad F_E = qE \quad E = \frac{V}{d}$$

$$a = \frac{F_E}{m} = \frac{qE}{m} = \frac{qV}{dm}$$

- F_E Electrostatic force
- m Mass
- a Acceleration
- q Charge
- E Electric field
- V Voltage
- d Distance

$d_1 > d_2 \quad a_1 = \frac{qV}{d_1 m} \quad a_2 = \frac{qV}{d_2 m} \quad a_2 > a_1$

Decreasing anode/cathode distance increases acceleration

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During electrophoresis, the **electrostatic force** F_E is equal to the product of the electric charge q of the protein and the magnitude of the **electric field** E :

$$F_E = qE$$

According to Newton's second law of motion, F_E is also equal to the product of the mass m of the protein and the **acceleration** a of the protein:

$$F_E = ma$$

Combining these equations and solving for a yields:

$$a = \frac{qE}{m}$$

Furthermore, the electrophoresis device generates an **electric field** E with a magnitude equal to the ratio of the **voltage** V and the distance d between the electrodes:

$$E = \frac{V}{d}$$

Substituting the value for E into the equation above for a gives:

$$a = \frac{qV}{dm}$$

Therefore, the protein experiences a constant a during electrophoresis that is directly proportional to q and V but **inversely proportional** to d and m .

In this question, the researcher wants to increase the value of a of the protein during electrophoresis. The above equation implies that decreasing d causes a to increase.

(Choice A) The number of protein molecules in the sample does not affect the value of a of an individual protein molecule. The value of a depends on the values of q and m for the protein molecule itself, but these values do not change when the number of protein molecules in the sample increases.

(Choice B) Increasing the distance d will decrease the value of a for the protein.

(Choice D) Decreasing the voltage V will decrease the value of a for the protein.

Educational objective:

The acceleration of a protein during electrophoresis depends on the charge of the protein, the voltage, the anode/cathode distance, and the mass of the protein. Decreasing the anode/cathode distance increases the acceleration of the protein.

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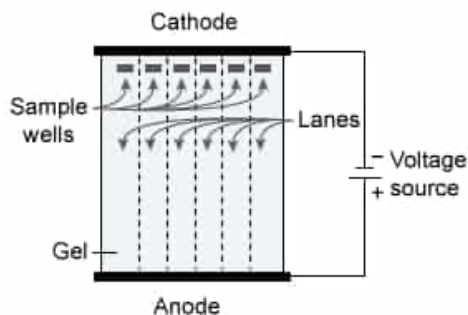


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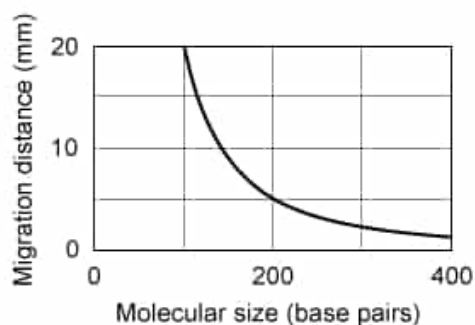


Figure 2 Plot of migration distance for molecules with different sizes

After sufficient time has elapsed for clear separation of the migration distances for each sample, the sample positions in the gel are detected by applying a stain or by tagging the molecules with markers that produce fluorescent or radioactive signals.

An electrophoresis device behaves like a 0.4 mF capacitor. What voltage results in a charge of 1 C on the anode?

- ✓ ☒ A. 2,500 V [57%]
☐ B. 400 V [18%]
☐ C. 250 V [16%]
☐ D. 2.5 V [8%]

Correct

57% answered correctly

Explanation:

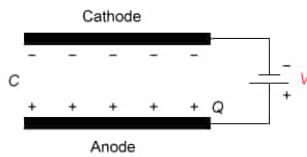
Voltage across electrophoresis anode and cathode

$$V = \frac{Q}{C}$$

- V Voltage
- Q Electric charge
- C Capacitance

$$C = 0.4 \text{ mF}$$

$$Q = 1 \text{ C}$$



$$V = \frac{1 \text{ C}}{0.4 \times 10^{-3} \text{ F}} = 2500 \text{ V}$$

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The **charge** Q on a capacitor equals the product of the capacitor's **capacitance** C and the **voltage** V :

$$Q = CV$$

This equation can be rearranged to calculate V as the ratio of Q and C :

$$V = \frac{Q}{C}$$

In this question, the electrophoresis device behaves like a capacitor with $C = 0.4 \text{ mF}$ and $Q = 1 \text{ C}$ on the anode. Therefore, V is given by:

$$V = \frac{1 \text{ C}}{0.4 \times 10^{-3} \text{ F}}$$

$$V = 2,500 \text{ V}$$

(Choice B) 400 V is obtained by incorrectly converting 0.4 mF to 400 F and multiplying Q and C .

(Choice C) 250 V is obtained by incorrectly using 4 mF for C .

(Choice D) 2.5 V is obtained by incorrectly using 0.4 F for C .

Educational objective:

An electrophoresis device can act as a capacitor. The voltage is equal to the ratio of the capacitance and the charge on the anode.

Time spent: 0 sec

Subject:
Physics

QID: 403463

System:
Fluids